Antioxidant status and its association with elevated depressive symptoms among US adults: National Health and Nutrition Examination Surveys 2005–06

We examined the relationship of elevated depressive symptoms with antioxidant status. Cross-sectional data from the National Health and Nutrition Examination Surveys 2005–06 on US adults aged 20–85 years were analyzed. Depressive symptoms were measured using the Patient Health Questionnaire with a score cutpoint of 10 to define "elevated depressive symptoms". Serum antioxidant status was measured by serum levels of carotenoids, retinol (free and retinyl esters), vitamin C and vitamin E. The main analyses consisted of multiple logistic and zero-inflated poisson regression models, taking into account sampling design complexity. The final sample consisted of 1,798 US adults with complete data. Higher total carotenoid serum level was associated with lower likelihood of elevated depressive symptoms with a reduction in the odds by 37% overall with each SD increase in exposure, and by 34% among women ($p<0.05$). A dose-response relationship was observed when serum total carotenoids were expressed as quartiles [Q4 (1.62–10.1 μmol/L) vs. Q1(0.06–0.86 μmol/L): OR=0.41; 95% CI: 0.23–0.76, $P<0.001$; p-value for trend=0.035], though no significant associations were found with other antioxidant levels. Among carotenoids, β-carotene (men and women combined) and lutein+zeaxanthins (women only, after control for dietary lutein+zeaxanthin intake and supplement use) had an independent inverse association with elevated depressive symptoms among US adults. None of the other serum antioxidants had a significant association with depressive symptoms, independently of total carotenoids and other covariates. In conclusion, total carotenoids (mainly β-carotene and lutein+zeaxanthins) in serum were associated with reduced levels of depressive symptoms among community-dwelling US adults.